Strong avoidance does not force local principal ideal domains

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Abstract

Every commutative ring with linearly ordered ideals has strong avoidance in the sense of Chen and Tarizadeh. This gives a four-element counterexample to their Conjecture 4.10. An explicit valuation domain with a nonprincipal maximal ideal also disproves the conjecture within integral domains.

Task 1. Resolve the strong-avoidance characterization. We disprove the characterization in Conjecture 4.10 of [1], using its actual Definition 4.2. All rings here are commutative with identity $1 \neq 0$, and ring maps preserve the identity. For a ring map $\varphi : R \to S$, write IS for the ideal generated by $\varphi(I)$. The ring S has *strong avoidance* when, for every such map and every finite nonempty family of ideals $I, I_1, \dots, I_n$ of R,

$$I \subseteq \bigcup_{j=1}^{n} I_j \quad \Longrightarrow \quad IS \subseteq I_j S \text{ for some } j.$$

The conjecture asserts that this is equivalent to $S_{\mathfrak{p}}$ being a principal ideal domain (PID) for every prime ideal $\mathfrak{p}$ of S. No Noetherian or reduced hypothesis appears in the source.

The MathDB paraphrase instead describes a condition on two ideals whose union is the whole ring. That condition is automatic: one of those ideals contains 1. Our counterexamples satisfy the stronger, actual source definition.

Lemma 1. *If the ideals of S are linearly ordered by inclusion, then S has strong avoidance.*

Proof. Fix an arbitrary map $\varphi : R \to S$ and a cover as above. Among the finite family of ideals $I_1S, \dots, I_nS$, choose a largest one, say I_jS. Then

$$\varphi(I) \subseteq \bigcup_{k=1}^{n} \varphi(I_k) \subseteq \bigcup_{k=1}^{n} I_k S = I_j S.$$

Since I_jS is an ideal, it contains the ideal generated by $\varphi(I)$, namely IS. The map need not be injective or surjective, and I need not be finitely generated. □

Proposition 1. *The ring $S = \mathbb{Z}/4\mathbb{Z}$ has strong avoidance, but its localization at its unique prime ideal is not a PID.*

Proof. Its ideals are (0), (2) and S, so Lemma 1 applies. Any prime ideal contains the nilpotent element 2; the quotient by (2) is $\mathbb{F}_2$, so (2) is the unique prime ideal. Every element outside (2) is a unit: 1 and 3 are their own inverses. Thus $S_{(2)} = S$. The element 2 is nonzero and has square zero, so S is not an integral domain and therefore is not a PID. □

The same argument applies to $\mathbb{F}_2[\epsilon]/(\epsilon^2)$, whose ideals are (0), (ϵ) and the whole ring. The next example shows that the obstruction persists even for domains, and even if "PID" were replaced by "principal ideal ring."

Proposition 2. *There is a valuation domain V with strong avoidance whose maximal ideal $\mathfrak{m}$ is not principal. In particular, $V_{\mathfrak{m}} = V$ is not a PID.*

Proof. Let t_0 be transcendental over $\mathbb{F}_2$, and in an algebraic closure of $\mathbb{F}_2(t_0)$ choose elements t_n with $t_{n+1}^2 = t_n$ for $n \geq 0$. Set

$$K = \bigcup_{n\geq 0} \mathbb{F}_2(t_n), \qquad V = \bigcup_{n\geq 0} \mathbb{F}_2[t_n]_{(t_n)}.$$

Both unions are nested. For the second, a polynomial denominator with nonzero constant term retains that property after substituting $t_n = t_{n+1}^2$. In particular, K is a field and V is a subring of K.

On $\mathbb{F}_2(t_n)^\times$ define

$$v(f) = 2^{-n} \operatorname{ord}_{t_n}(f).$$

Substitution doubles the order of every nonzero rational function, so these definitions agree on overlaps and define a valuation $v : K^\times \to \mathbb{Z}[1/2]$. Every dyadic rational occurs, since $v(t_n^a) = a/2^n$ for $a \in \mathbb{Z}$. At each finite stage, nonnegative order is exactly the condition for membership in its local polynomial ring. Consequently

$$V = \{0\} \cup \{f \in K^\times : v(f) \geq 0\}.$$

For every nonzero $f \in K$, either f or f^{-1} belongs to V.

To check that all ideals of V are comparable, suppose $A \not\subseteq B$ and choose $a \in A \setminus B$. For nonzero $b \in B$, the element a/b cannot belong to V, since that would imply $a \in B$. Hence $b/a \in V$ and $b \in A$. Thus $B \subseteq A$. Lemma 1 now proves strong avoidance for V.

The nonunits of V form the ideal

$$\mathfrak{m} = \{0\} \cup \{f \in K^\times : v(f) > 0\} :$$

the valuation inequality shows closure under addition, and multiplication by elements of V preserves positivity. Elements of value zero have their inverses in V. Thus $\mathfrak{m}$ is the unique maximal ideal, and localizing there changes nothing.

If $\mathfrak{m} = (a)$, then $a \neq 0$ and $q = v(a) > 0$. Write $q = c/2^n$ with integers $c > 0$ and $n \geq 0$. The element $b = t_{n+1}^c$ has value $q/2 > 0$, so $b \in \mathfrak{m}$. But $v(b/a) = -q/2 < 0$, whence $b/a \notin V$ and $b \notin (a)$, a contradiction. Therefore $\mathfrak{m}$ is not principal. □

Either proposition disproves the necessary direction of the stated equivalence. Proposition 2 does so within integral domains. These counterexamples do not assert a complete replacement classification.

Task 2. Classify finite local examples. Determine which finite commutative local rings have strong avoidance. In particular, decide whether every such ring must have its ideals linearly ordered, and identify the role of the residue field and nilpotent ideals.

Task 3. Determine the effect of finite products. Find necessary and sufficient conditions for a finite product of rings with strong avoidance to retain that property. Track when the containing ideal can be chosen with the same index in every factor.

Task 4. Investigate the domain classification. Determine which Prüfer domains have strong avoidance, and whether every domain with strong avoidance must be Prüfer. The valuation-domain lemma settles the local sufficiency direction but leaves the global choice of an index unresolved.

Task 5. Identify valid preservation statements. Determine when strong avoidance passes to localizations, quotients and directed unions. Use those results to formulate and test a corrected characterization under explicit finiteness or local hypotheses.

References

[1] J. Chen and A. Tarizadeh, *On the ideal avoidance property*, arXiv:2203.04256v4 (6 September 2023), Definition 4.2, p. 12, and Conjecture 4.10, p. 14. Journal reference: J. Pure Appl. Algebra **228**(3) (2024), 107500, doi:10.1016/j.jpaa.2023.107500. The proof uses the current author-linked arXiv version; the final typeset journal text was unavailable during the audit.